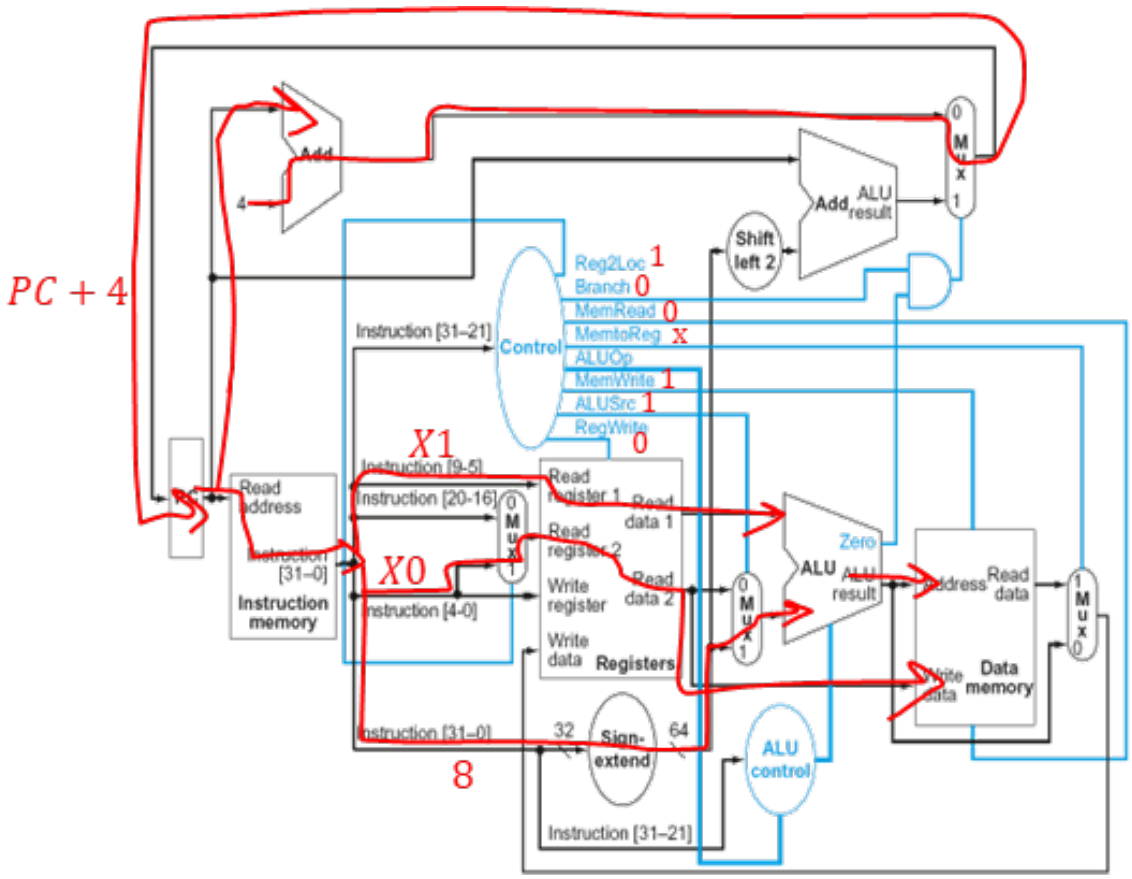
CSCE 2610 – Spring 2026 – Computer Organization Name: _____

In-Class Activity #4

20 Points

1. Trace the datapath for the STUR X0, [X1, #8]. **(10 Points)**



STUR X0, [X1, #8]

31 – 21	20 – 12	11 – 10	9 – 5	4 – 0
opcode	address	op2	Rn	Rt
11 bits	9 bits	2 bits	5 bits	5 bits

2. Identify all the stalls and the total number of stalls in the assembly code. Assume there is no forwarding and no branch prediction. **(5 points)**

LSL X10, X19, #1

LSL X10, X10, #3

ADD X10, X10, X22

ADD X9, X20, X21

STUR X9, [X10, #0]

ADDI X20, X20, #2

STUR X20, [X22, #0]

LSL X10, X19, #1

2 *stalls*

LSL X10, X10, #3

2 *stalls*

ADD X10, X10, X22

ADD X9, X20, X21

2 *stalls*

STUR X9, [X10, #0]

ADDI X20, X20, #2

2 *stalls*

STUR X20, [X22, #0]

8 stalls in total

3. Reorder the above code to reduce the stalls without changing functionality. Assume there is no forwarding and no branch prediction. **(5 points)**

```
LSL X10, X19, #1
ADD X9, X20, X21
ADDI X20, X20, #2
LSL X10, X10, #3
ADD X10, X10, X22
STUR X20, [X22, #0]
STUR X9, [X10, #0]
```

2 stalls

1 stall

3 stalls in total